

UTKLIPPAN, Sweden

WMO: 026320

Lat: 55.953N

Lon: 15.703E

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.6	-4.3	-9.4	1.7	-4.1	-7.7	2.0	-2.5	17.5	3.3	16.4	2.6	7.7	20	0.790

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.6	23.0	19.9	21.8	19.2	20.9	18.5	20.7	22.3	19.7	21.3	18.8	20.4	4.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	14.8	22.0	19.0	13.8	20.7	18.1	13.0	20.0	60.0	22.3	56.2	21.4	53.5	20.5	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.6	14.2	13.0	-6.9	23.6	2.7	1.6	-8.9	24.8	-10.5	25.7	-12.0	26.5	-14.0	27.7		
(5)			-7.0	20.8	2.5	1.8	-8.7	22.1	-10.2	23.1	-11.6	24.1	-13.4	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	1.3	1.2	2.3	5.2	10.0	14.7	17.6	18.1	14.4	10.2	6.5	3.5
	DBStd	6.49	2.65	2.68	2.33	1.70	2.25	2.08	2.39	2.06	2.28	2.56	2.50	3.16
	HDD10.0	1264	271	246	239	143	29	0	0	0	1	29	106	201
	HDD18.3	3524	529	479	497	393	259	109	43	29	119	253	354	459
	CDD10.0	826	0	0	0	0	28	142	236	250	133	35	2	0
	CDD18.3	44	0	0	0	0	0	1	21	21	2	0	0	0
	CDH23.3	28	0	0	0	0	0	0	17	12	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	7.0	8.1	8.0	7.1	6.2	6.0	5.9	5.6	6.2	7.0	7.7	8.1	8.6
Precipitation	PrecAvg	571	42	38	34	28	39	42	48	63	60	61	61	53
	PrecMax	717	86	76	75	54	107	107	149	200	156	154	159	125
	PrecMin	423	2	7	5	1	11	0	1	12	21	13	28	24
	PrecStd	84	22	15	18	15	23	26	31	44	33	34	30	23
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.5	6.4	7.7	10.8	16.5	21.4	25.0	24.8	21.2	15.9	12.5	9.4
		MCWB	5.7	4.9	5.7	8.1	13.8	18.0	21.4	21.3	17.8	14.5	11.5	8.5
	2%	DB	5.8	5.6	6.3	9.2	15.3	19.8	23.5	22.7	19.7	14.8	11.4	8.7
		MCWB	5.1	4.6	4.8	7.1	12.9	17.1	20.5	19.4	16.8	13.7	10.6	7.9
	5%	DB	5.2	5.1	5.6	8.4	14.2	18.7	22.2	21.8	18.4	14.1	10.4	8.0
		MCWB	4.5	4.3	4.4	6.6	12.0	16.3	19.7	19.0	16.1	13.0	9.6	7.3
	10%	DB	4.7	4.6	5.0	7.5	13.2	17.7	21.2	20.9	17.4	13.5	9.4	7.3
		MCWB	4.1	3.8	4.0	6.1	11.3	15.7	19.0	18.6	15.4	12.5	8.6	6.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.2	5.6	6.1	8.5	14.5	18.8	22.2	21.7	18.6	15.0	12.0	8.8
		MCDB	6.3	5.9	7.1	9.9	16.0	20.2	24.0	23.4	20.2	15.4	12.4	9.1
	2%	WB	5.3	4.9	5.2	7.6	13.3	17.6	21.1	20.6	17.5	14.0	10.7	8.1
		MCDB	5.6	5.4	6.0	8.8	14.8	19.1	22.8	22.2	19.0	14.5	11.2	8.6
	5%	WB	4.8	4.4	4.7	6.9	12.4	16.7	20.2	19.4	16.6	13.2	9.8	7.5
		MCDB	5.2	5.0	5.4	8.1	13.9	18.2	21.8	21.1	17.9	13.8	10.4	8.0
	10%	WB	4.2	4.0	4.2	6.2	11.5	16.0	19.2	18.8	15.7	12.6	8.8	6.7
		MCDB	4.7	4.6	4.9	7.3	12.9	17.4	21.0	20.4	17.0	13.3	9.3	7.2
Mean Daily Temperature Range	5% DB	MDBR	2.5	2.4	2.8	3.2	3.5	3.9	3.8	3.6	3.2	2.6	2.3	2.4
		MCDBR	2.3	2.4	3.5	4.3	4.4	4.9	4.8	4.0	3.9	3.0	2.3	2.0
		MCWBR	2.4	2.1	2.7	2.9	2.8	3.1	2.8	2.8	2.7	2.4	2.6	2.1
	5% WB	MCDBR	2.3	2.3	3.2	4.1	4.0	4.4	4.2	3.5	3.4	2.5	2.3	1.9
		MCWBR	2.6	2.1	2.6	2.9	2.8	3.0	2.7	2.7	2.7	2.4	2.6	2.1
Clear-Sky Solar Irradiance	taub	0.315	0.341	0.359	0.391	0.389	0.380	0.406	0.405	0.375	0.361	0.343	0.308	
	taud	2.294	2.314	2.322	2.303	2.380	2.438	2.386	2.385	2.444	2.390	2.310	2.262	
	Ebn at Noon	586	717	801	826	853	864	831	805	781	686	541	492	
	Edh at Noon	54	76	96	112	110	106	109	103	85	71	53	44	
All-Sky Solar Radiation	RadAvg	0.41	1.02	2.47	4.25	5.61	6.01	5.46	4.41	3.06	1.44	0.55	0.29	
	RadStd	0.07	0.19	0.23	0.47	0.53	0.37	0.60	0.36	0.33	0.24	0.10	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (42 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-155	+60	N/A

Nomenclature: See separate page